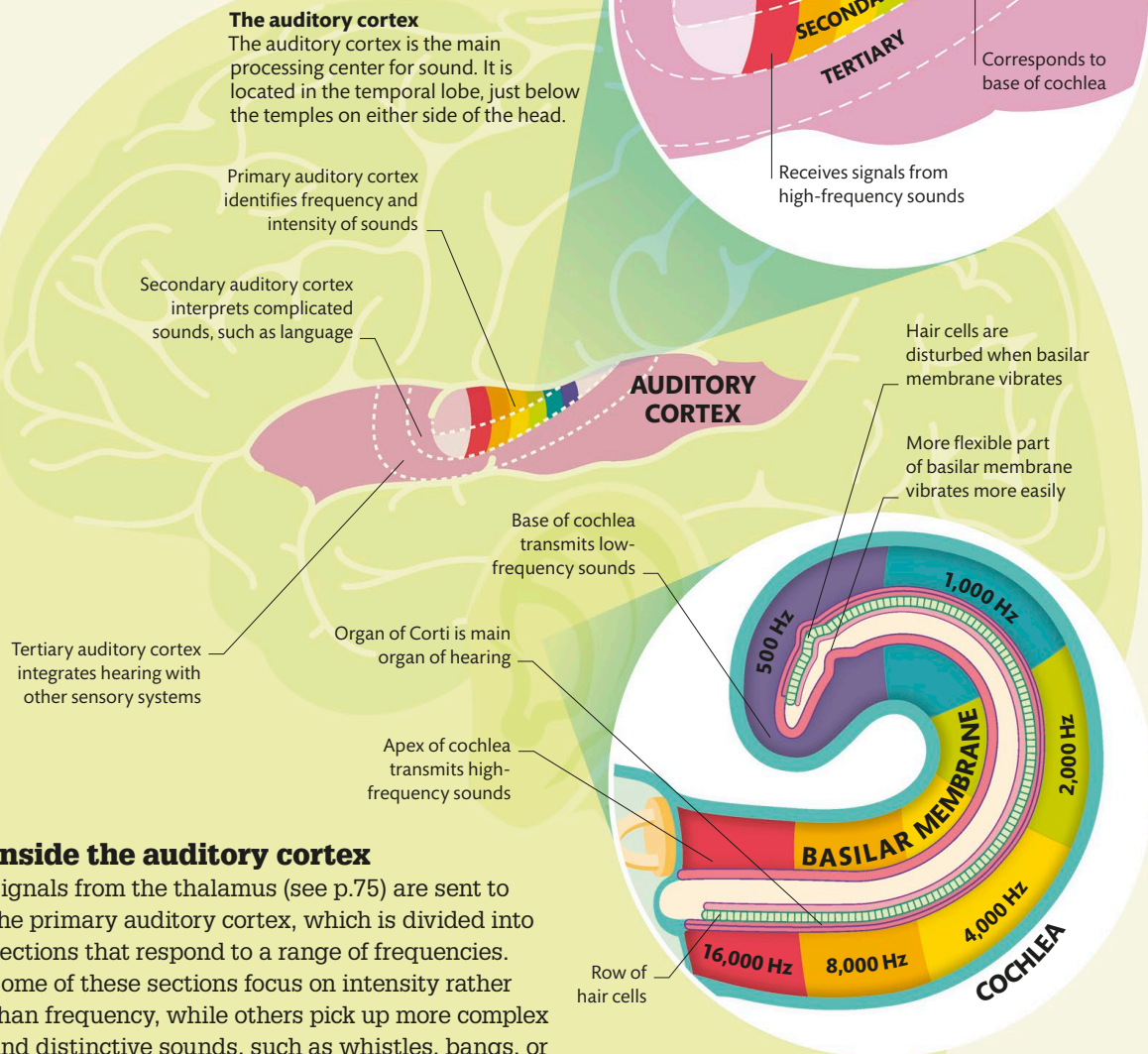


Perceiving Sound

Every sound is made up of a number of different components. The brain has to take all the details of its frequencies, intensity, and rhythm to process, identify, and remember the sound.



Inside the auditory cortex

Signals from the thalamus (see p.75) are sent to the primary auditory cortex, which is divided into sections that respond to a range of frequencies. Some of these sections focus on intensity rather than frequency, while others pick up more complex and distinctive sounds, such as whistles, bangs, or animal noises. Signals then pass to the secondary auditory cortex, which is thought to focus on harmony, rhythm, and melody. The tertiary auditory cortex integrates all the signals to give an overall impression of the sounds picked up by the ears.

The cochlea

Areas along the curl of the cochlea respond to different frequencies of sound, from high-pitched at the apex to low bass notes at the base. These are mirrored by corresponding areas in the auditory cortex.